MATH11218
Methods and Basically Specialist Revision

Authored by: Lachlan J. W. Wyer

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1 Introduction

1.1 Indices:

Within the expression of a^m, a is considered to be the base with the power m being the index. The reason that this is considered useful is in scientific notation to expression large numbers, or simplify repeated multiplication, fractions, roots or polynomials. E.g. $2 \times 2 \times 2 \times 2 \times 2 \times 2 = 2^6$. With these there are a number of different index laws that goes with it:

1. $a^m \cdot a^n = a^{m+n}$
2. $\frac{a^m}{a^n} = a^{m-n}$
3. $(a^m)^n = a^{m \cdot n}$
4. $(ab)^m = a^m b^m$
5. $(\frac{a}{b})^m = \frac{a^m}{b^m}$
6. $a^{\frac{m}{n}} = \sqrt[n]{a^m}$
7. $a^0 = 1$

1.2 Poynomials:

A polynomial explains/denotes the degree or curviness that a line is at. Such as a constant can be described as being degree 0, linear degree 1, quadratic degree 2, cubic degree 3 and so on.

1.2.1 Quadratics

With quadratics it is important to understand that it can contain multiple different types of roots. For example, iff: $\Delta = b^2 - 4ac$

- $\Delta > 0$, when it contains 2 real roots, e.g. $(x + a)(x + b)$.
- $\Delta = 0$, when there is 1 real root, e.g. $(x + a)^2$.
- $\Delta < 0$, when no real roots are found, e.g $(x + aj)(x + bj)$.

1.2.2 Polynomials with Degrees Greater than 2

When a polynomial contains a degree that is greater than 2, it roots can be found through a process called polynomial long division.

$$
\begin{array}{r|l}
x+2 & \begin{array}{r} x^2 + x - 12 \\ \hline x^3 + 3x^2 - 10x - 24 \\ -x^3 - 2x^2 \\ \hline x^2 - 10x \\ -x^2 - 2x \\ \hline -12x - 24 \\ 12x + 24 \\ \hline 0 \end{array}
\end{array}
$$

$$
\begin{aligned}
\therefore\ x^3 + 3x^2 - 10x - 24 &= (x+2)(x^2 - x - 12) \\
&= (x+2)(x-4)(x+3) \\
x &= -3,\ -2,\ 4
\end{aligned}
$$

1.3 Algebraic Functions:

Within fractional functions there are two main types:

1. Proper fractions, where the denominator is smaller then the numerator.
2. Improper fractions, where the denominator is less or equal to the numerator.

It is important to note, that the greater, equal to, and less than are only relative to polynomial degree. Some parts to with division of fraction is just the multiplication of the recipricol, $\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \times \frac{d}{c}$.

$$
\begin{aligned}
\frac{2}{x+1} + \frac{4}{x+2} &= 0 \\
&= \frac{2(x+2) + 4(x+1)}{(x+1)(x+2)} \\
&= \frac{2x + 4 + 4x + 4}{(x+1)(x+2)} \\
&= \frac{6x + 8}{(x+1)(x+2)} \\
\therefore\ x &\neq -2,\ -1
\end{aligned}
$$

2 Sum Notation

2.1 Inequalities:

$<$ - Less than

$>$ - Greater than

$\leq$ - Less than or equal to

$\geq$ - Greater than or equal to

For example:

$$\begin{aligned} 1 - 2t &< 7 \\ -2t &< 6 \\ t &> -3 \end{aligned}$$

$\frac{x}{y} > 0$ OR $xy > 0$, presents two different cases in the possible solutions that it presents:

Case i - $x \,\&\, y > 0$

Case ii - $x \,\&\, y < 0$

$\frac{x}{y} < 0$ OR $xy < 0$, also presents two different cases in the possible solutions that it presents:

Case i - $x < 0 \& y > 0$

Case ii - $x > 0 \& y < 0$

2.2 Partial Fractions:

These are used on improper fractions to complete polynomial long division.

$$\begin{array}{r} x^2 + x + 3 \\ 2x-2\,\overline{)\;2x^3 \quad + 4x - 1} \\ \underline{-2x^3 + 2x^2} \\ 2x^2 + 4x \\ \underline{-2x^2 + 2x} \\ 6x - 1 \\ \underline{-6x + 6} \\ 5 \end{array}$$

$$2x^3 + 4x - 1 = x^2 + x + 3 + \frac{5}{2x-2}$$

Some equations for the partial fractions, always assuming that the denominator is of a higher degree than the numerator:

1. $\frac{f(x)}{(x+a)(x+b)} = \frac{A}{x+a} + \frac{B}{x+b}$
2. $\frac{g(x)}{(x+c)^n} = \frac{A}{x+c} + \frac{B}{(x+c)^2} + \ldots + \frac{Z}{(x+c)^n}$
3. $\frac{h(x)}{(ax^2+bx+c)(x+d)} = \frac{Ax+B}{ax^2+bx+c} + \frac{C}{x+d}$

(e.g.)

$$\frac{2x^4 - 8x2 + 5x - 2}{x^3 - 4x}$$

$$x^3 - 4x \overline{)\; 2x^4 - 8x^2 + 5x - 2} \quad 2x$$

$$\underline{-2x^4 + 8x^2}$$

$$5x - 2$$

$$\frac{2x^4 - 8x2 + 5x - 2}{x^3 - 4x} = 2x + \frac{5x - 2}{x^3 - 4x}$$

$$\begin{aligned} \frac{5x-2}{x(x^2-4)} &= \frac{A}{x} + \frac{Bx+C}{x^2-4} \\ &= \frac{A(x^2-4) + (Bx+C)x}{x(x^2-4)} \end{aligned}$$

$$\begin{aligned} \therefore\ 5x - 2 &= A(x^2-4) + (Bx+C)x \\ &= Ax^2 - 4A + Bx^2 + Cx \end{aligned}$$

x^2	x^1	x^0
$0 = A + B$	$5 = C$	$-2 = -4A$
$0 = \frac{1}{2} + B$		$A = \frac{1}{2}$
$B = -\frac{1}{2}$		

3 Vectors

3.1 Polar Co-ordinates:

Here just consider the "x" and "y" co-ordinates of a vector as just the relationship between the angle and hypotenuse.

$$r < \theta$$

Where:

- r is the magnitude
- θ is the angle, usually measured from the x-axis

As such, to find the different aspects of polar coordinates use the following formulas.

$$x = r\cos(\theta)$$
$$y = r\sin(\theta)$$

$$r = \sqrt{x^2 + y^2}$$
$$\theta = \arctan(\frac{y}{x})$$

For 3D vectors:

$$r < \theta < \varphi$$

3.2 Vectors:

Remember the scene from Minions, Vector(s) need both direction & and magnitude, $\hat{v} = \frac{\vec{v}}{|v|}$.

4 Engineering Functions I

4.1 Number Systems:

Within the part of pure mathematics there are several different types of number systems:

- $\mathbb{N}$: All types of numbers. This can be further split into $\mathbb{N}^+$ & $\mathbb{N}^-$
- $\mathbb{W}$: Just the whole numbers
- $\mathbb{Z}$: The interger set
- $\mathbb{Q}$: Rational numbers, these include numbers that terminate, recurring of can be defined in a fraction
- $\mathbb{I}$: Irrational numbers such as π, e, or $\sqrt{2}$
- $\mathbb{R}$: Real numbers
- $\mathbb{C}$: Complex set

4.2 The Function:

The function f(x), define the range [f(x)] as dependend on the domain [x]. This allows the function be restricted over a specified domain. Within functions there are several main types of functions. For a graph to be considered a function it must pass the vertical line test.

4.2.1 Even Function

These are functions that are symmetrical about the yaxis, e.g. $f(-x) = f(x)$.

4.2.2 Odd Function

These are symmetrical about the rotation of the origin, e.g. $f(-x) = -f(x)$.

4.2.3 Domain

The domain can be specified as the set of values that can be inputed into the function. In real-world examples, physical properties, generally cannot be negative, and as such a restricted domain set must be used.

4.2.4 Range

This is just the output of the restricted set of the domain, based upon the actions that the function takes.

4.3 Composite Functions:

E.g. $f[g(x)]$. Here the range of the inner function affects the domain of the outer function.

$$\begin{aligned} f(x) &= 4x + 1 \\ g(x) &= 2x^2 + 5x \end{aligned}$$

$$\begin{aligned} g[f(x)] &= 2(4x + 1)^2 + 5(4x + 1) \\ &= 2(16x^2 + 8x + 1) + (20x + 5) \\ &= 32x^2 + 36x + 7 \end{aligned}$$

$$\begin{aligned} f[g(x)] &= 4(2x^2 + 5x) + 1 \\ &= 18x^2 + 20x + 1 \end{aligned}$$

5 Engineering Functions II

5.1 Parametric Functions:

An x and y function are both affected by third variable such as distance, time or angle. For example:

$$\begin{aligned} x &= t^2 \\ \pm\sqrt{x} &= 2 \\ y &= 2t \end{aligned}$$

$$\begin{aligned} \therefore\ y &= \pm 2\sqrt{x} \\ y &= \sqrt{2},\ \leq t \leq 5 \end{aligned}$$

5.2 Inverse Functions:

$y = f^{-1}(x)$ is just the line $y = f(x)$ reflected over $y = x$. An example of this is cos and arccos, e and ln().

5.3 Break Functions:

5.3.1 Continuous Functions

A function in where there is no break in the continuation of the function.

5.3.2 Piecewise Functions

The function contains breaks. These are points where the function might not be defined or allow joining multiple function together.

5.3.3 Peirodic Functions

This is type of function that repeats itself after a certain period. A famouns example is trigonometric functions.

5.3.4 Rational Functions

These functions are represented as fractions, $R(x) = \frac{P(x)}{Q(x)}$. It is key to note that these functions contain at least one asymptote. With the function below,

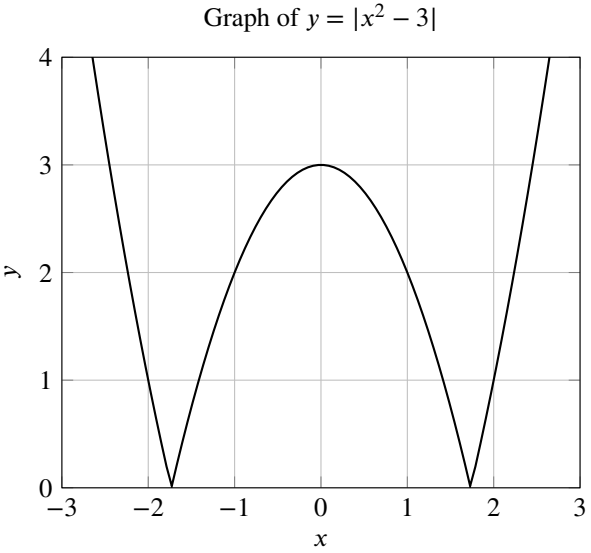
@ x=-2 and x=1 it will be undefined.

$$\begin{aligned} y &= \frac{x}{x^2 + x - 2} \\ &= \frac{x}{(x+2)(x-1)} \end{aligned}$$

$$x = -2,\ 1$$

5.3.5 Modulus Function

This is a function where the range is always positive.

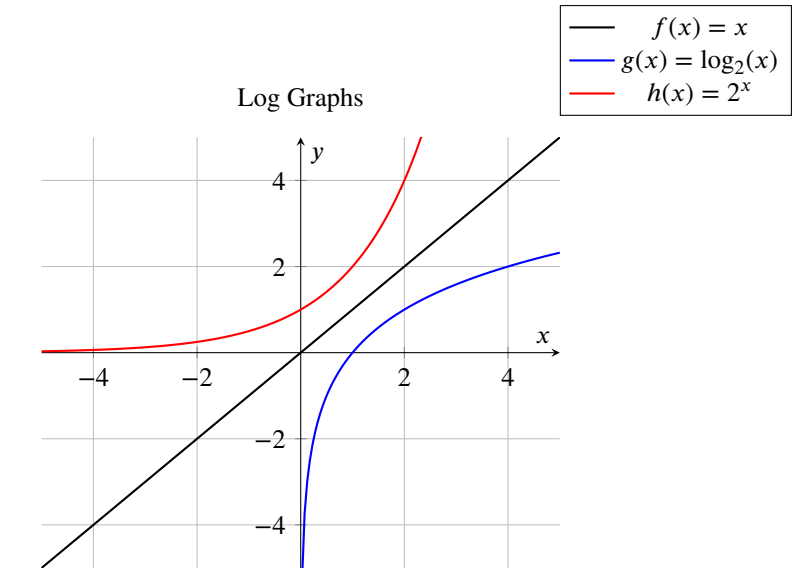


5.4 Logarithms:

5.4.1 Graphing

Log functions are the inverse of the exopential.

$$a^x = b,\ log_a(b) = x$$

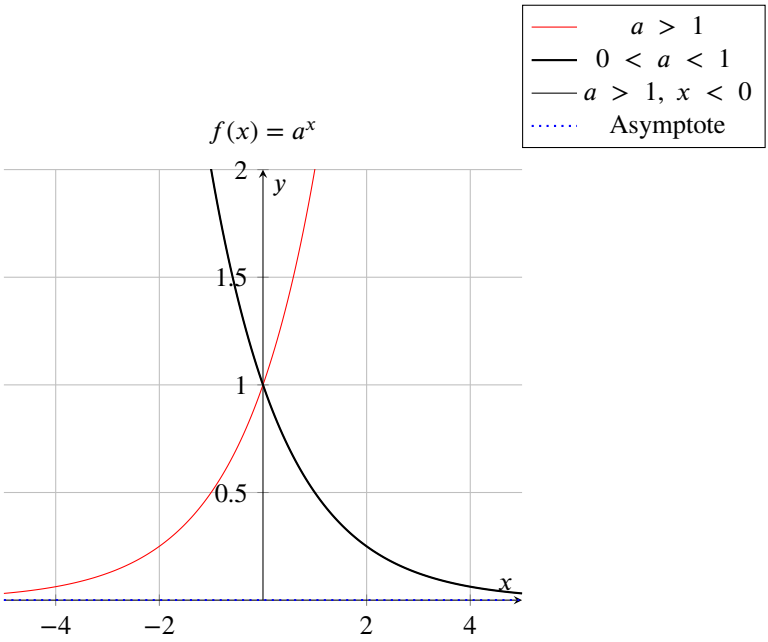


5.4.2 Log Laws

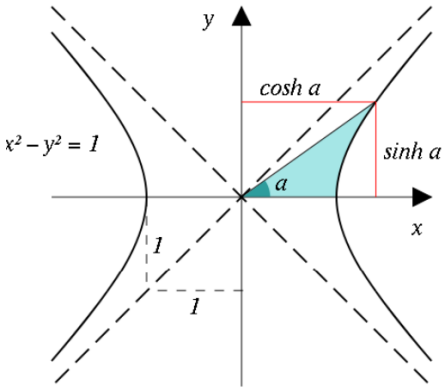
$$\log_x(a) + \log_x(b) = \log_x(ab)$$
$$\log_x(a) + \log_x(b) = \log_x(\frac{a}{b})$$
$$\log_x(a^m) = m\log_x(a)$$
$$\log_a(a) = 1$$
$$\log_x(1) = 0$$
$$\log_a(b) = \frac{\log_x(b)}{\log_x(a)}$$

There are also some interesting properties of logs in functions, such as $a^{\log_a(x)} = x$ as well as $e^{\ln(x)} = x$.

5.5 Exponential Functions:



5.6 Hyperbolic Functions:



$$\sinh(x) = \frac{e^x - e^{-x}}{2} \qquad \operatorname{csch}(x) = \frac{2}{e^x - e^{-x}}$$
$$\cosh(x) = \frac{e^x + e^{-x}}{2} \qquad \operatorname{sech}(x) = \frac{2}{e^x + e^{-x}}$$
$$\tanh(x) = \frac{\sinh(x)}{\cosh(x)} \qquad \coth(x) = \frac{e^x + e^{-x}}{e^x - e^{-x}}$$
$$= \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

5.6.1 Hyperbolic Identities

$$\sinh(-x) = -\sinh(x)$$
$$\cosh(-x) = \cosh(x)$$

$$\cosh^2(x) - \sinh^2(x) = 1$$
$$1 - \tanh^2(x) = \operatorname{sech}^2(x)$$
$$\coth^2(x) - 1 = \operatorname{csch}^2(x)$$

$$\sinh(x \pm y) = \sinh(x)\cosh(y) \pm \cosh(x)\sinh(y)$$
$$\cosh(x \pm y) = \cosh(x)\cosh(y) \pm \sinh(x)\sinh)y)$$

$$\sinh(2x) = 2\sinh(x)\cosh(x)$$
$$\cosh(2x) = \cosh^2(x) + \sinh^2(x)$$

$$2\cosh^2(x) = \cosh(2x) + 1$$
$$2\sinh^2(x) = \cosh(2x) - 1$$

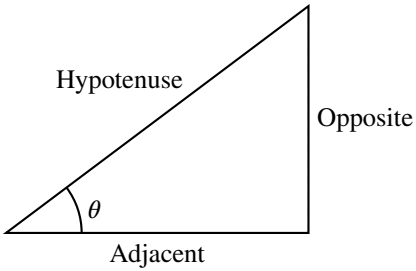
$$e^x = \cosh(x) + \sinh(x)$$
$$e^{-x} = \cosh(x) - \sinh(x)$$

6 Trignometry

6.1 Angles:

Positives angles rotate anti-clockwise, with negatives being in a clockwise direction. The two lengths that an angles rotates between is called the initial & terminal sides. When measuring angles on a cartesian plane, the standard position is the positive x-axis and rotating the standard anti-clockwise.

6.2 Trig Ratios:

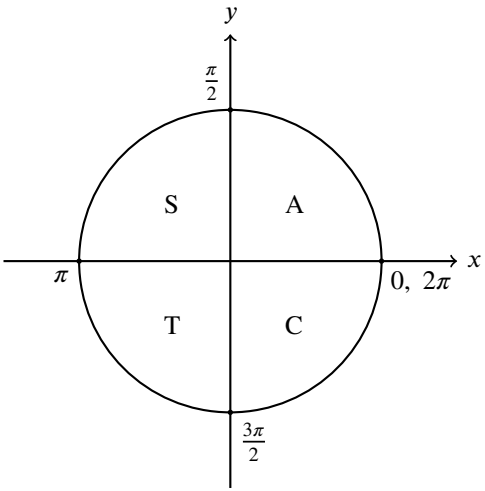


$$\sin(\theta) = \frac{O}{H} \qquad \cos(\theta) = \frac{A}{H} \qquad \tan(\theta) = \frac{O}{A}$$

$$\csc(\theta) = \frac{1}{\sin(\theta)} = \frac{H}{O} \qquad \sec(\theta) = \frac{1}{\cos(\theta)} = \frac{H}{A} \qquad \cot(\theta) = \frac{1}{\tan(\theta)} = \frac{A}{O}$$

To remember the ratios of trig functions to fractions of the sides remember the accroymn SOH CAH TOA.

6.3 Unit Circle:



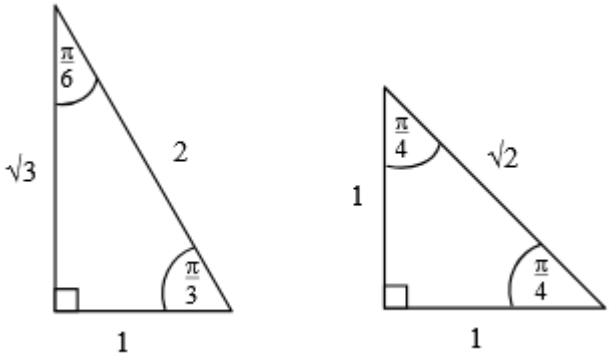
When dealing angle accuracy is key, so a system was developed to make angles more accurate. To do this fractions of π was used, known as radians.

$$D \to R = \theta \times \frac{\pi}{180}$$

$$R \to D = \phi \times \frac{180}{\pi}$$

6.4 Exact Values:

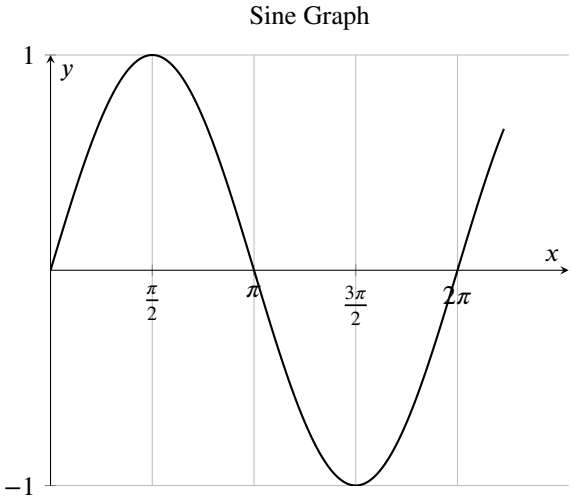
6.4.1 Triangles



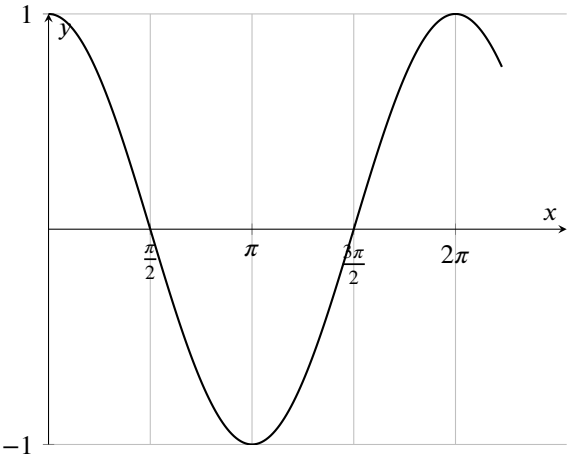
6.4.2 Table

Angle	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	π
$\sin(\theta)$	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1	0
$\cos(\theta)$	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0	-1
$\tan(\theta)$	0	$\frac{\sqrt{3}}{3}$	1	$\sqrt{3}$	undef	0
$\csc(\theta)$	undef	2	$\sqrt{2}$	$\frac{2\sqrt{3}}{3}$	1	undef
$\sec(\theta)$	1	$\frac{2\sqrt{3}}{3}$	$\sqrt{2}$	2	undef	-1
$\cot(\theta)$	0	$\sqrt{3}$	1	$\frac{\sqrt{3}}{3}$	0	undef

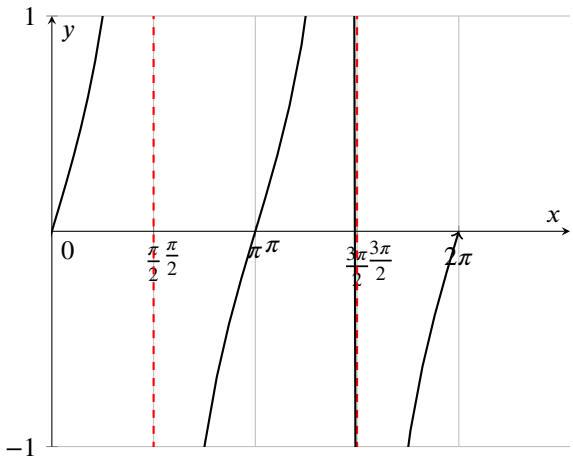
6.5 Graphing:



Cosine Graph



Tangent Graph



It is important to understand that 1 period in a tan/cot graph is $\frac{\pi}{B}$.

6.6 Quadrant System:

When using angles outside of the first quadrant, it is important to keep in mind the sign of the trig function (CAST system) and using the reference angle.

6.7 Trig Identities:

$$\tan(\theta) = \frac{\sin(\theta)}{\cos(\theta)}$$

$$\begin{aligned}
\cos^2(\theta) + \sin^2(\theta) &= 1 \\
\tan^2(\theta) + 1 &= \sec^2(\theta) \\
\cot^2(\theta) + 1 &= \csc^2(\theta)
\end{aligned}$$

$$\begin{aligned}
\sin(A \pm B) &= \sin(A)\cos(B) \pm \cos(A)\sin(B) \\
\cos(A \pm B) &= \cos(A)\cos(B) \mp \sin(A)\sin(B) \\
\tan(A \pm B) &= \frac{\tan(A) \pm \tan(B)}{1 \mp \tan(A)\tan(B)}
\end{aligned}$$

$$\begin{aligned}
\sin(2A) &= 2\sin(A)\cos(B) \\
\cos(2A) &= \cos^2(A) - \sin^2(A) \\
&= 1 - 2\sin^2(A) \\
&= 2\cos^2(A) \\
\tan(2A) &= \frac{2\tan(A)}{1 - \tan^2(A)}
\end{aligned}$$

$$\begin{aligned}
\sin(A) \pm \sin(B) &= 2\sin(\frac{A \pm B}{2})\cos(\frac{A \mp B}{2}) \\
\cos(A) + \cos(B) &= 2\cos(\frac{A + B}{2})\cos(\frac{A - B}{2} \\
\cos(A) - \cos(B) &= -2\sin(\frac{A + B}{2})\sin(\frac{A - B}{2})
\end{aligned}$$

Example of using these equations:

$$\frac{2\cos^2(x)-1}{\sin(x)\cos(x)} = \cot(x) - \tan(x)$$

$$\begin{aligned}
LHS &= \frac{2\cos^2(x)-1}{\sin(x)\cos(x)} \\
&= \frac{\cos(2x)}{\frac{1}{2}\sin(2x)} \\
&= 2 \times \frac{\cos(2x)}{\sin(x)} \\
&= 2 \times \cot(2x) \\
&= 2 \times \frac{1}{\tan(x)} \\
&= 2 \times \frac{1-\tan^2(x)}{2\tan(x)} \\
&= \frac{1}{\tan(x)} - \tan(x) \\
&= \cot(x) - \tan(x) \\
&= RHS
\end{aligned}$$

6.8 Modelling Waveforms:

$$y = A\sin(\omega t + \phi) \text{ OR } y = A\cos(\omega t + \phi)$$

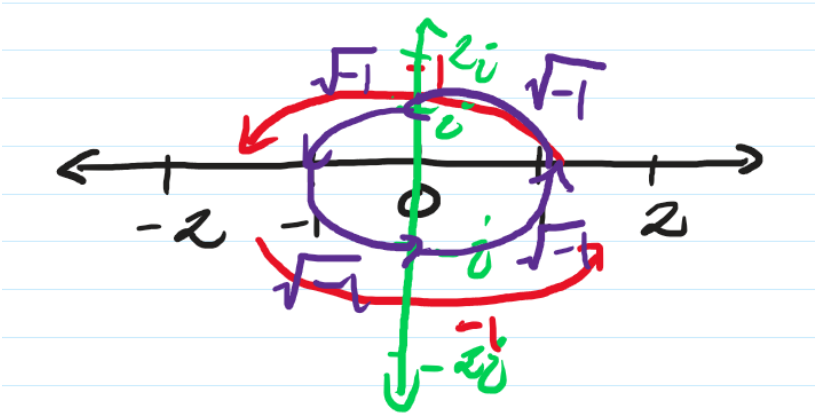
- A: Amplitude
- ω: Angular frequency (rad/sec)
- ϕ: Phase angle (rad)
- f: Frequency ($\frac{\omega}{2\pi}$)
- T: Period ($\frac{2\pi}{\omega}$ OR $\frac{1}{f}$)
- s: Time displacement ($\frac{\phi}{\omega}$)

To combine waveforms into one single waveform, expand the waveform function required with the appriotate trig identities.

7 Complex Numbers

7.1 The Basics:

By utilizing complex numbers, it allows mathematicians to complete $\sqrt{(}-x)$, where x > 0. As such the complex plane is just the standard number plane rotated 90°. In electrical engineering the complex numbers are represented by the letter j rather then the standard i.



7.2 Complex Number Forms:

7.2.1 Cartesian Form

This form represents a complex number as parts of the Re(z) and Im(z).

$$z = a + bj$$

With it there are two points to remember. Equality, when the Re=Re and Im=Im, and conjugates, the complex number is reflected over the x-axis.

7.2.2 Polar Form

$$\begin{aligned} z &= (a, b) \\ &= r < \theta \\ &= r\cos(\theta) + r\sin(\theta)j \end{aligned}$$

$$\begin{aligned} r &= a^2 + b^2 \\ \theta &= \arctan(\frac{b}{a}) \\ -\pi < \theta &\leq \pi \end{aligned}$$

7.2.3 Exponential Form

$$\begin{aligned} re^{j\theta} &= r\cos(\theta) + r\sin(\theta)j \\ re^{-j\theta} &= r\cos(\theta) - r\sin(\theta)j \end{aligned}$$

$$\begin{aligned} \cos(\theta) &= \frac{e^{j\theta} + e^{-j\theta}}{2} \\ \sin(\theta) &= \frac{e^{j\theta} - e^{-j\theta}}{2} \end{aligned}$$

$$\begin{aligned} \cos(j\theta) &= \cosh(\theta) \\ \sin(j\theta) &= \sinh(\theta) \end{aligned}$$

7.3 Operations on Complex Numbers:

7.3.1 Addition & Substraction

This can really only be done on complex numbers in cartesian form. Any complex numbers in other forms must first be convereted to this form first before doing the addition and subtraction.

7.3.2 Multiplication & Division

Whilst you can multiply complex numbers in cartesian form through the operation like Crab Claw, and division by multiplication of 1 through

conjutate($\bar{z}$) divide conjutate($\bar{z}$).

A much more simple method using the complex numbers in polar form.

$$z_1 = r_1 < \theta_1$$
$$z_2 = r_2 < \theta_2$$

$$z_1 z_2 = r_1 r_2 < (\theta_1 + \theta_2)$$
$$\frac{z_1}{z_2} = \frac{r_1}{r_2} < (\theta_1 - \theta_2)$$
$$z_1^n = r_1^n < n\theta_1$$
$$\bar{z_1} = r_1 < -\theta_1$$

7.4 De Moivre's Theorem:

This theorem allow the quick multiplication of the same complex number multiple times, such as having the number to a power.

$$\begin{aligned} z &= x + yj \\ &= r < \theta \\ &= r[\cos(\theta) + \sin(\theta)j] \\ &= re^{j\theta} \end{aligned}$$

$$(re^{j\theta})^n = r^n e^{jn\theta}$$
$$(r[\cos(\theta) + \sin(\theta)j])^n = r^n[\cos(n\theta) + \sin(n\theta)j]$$
$$(r < \theta)^n = r^n < n\theta$$

7.5 nth Roots of Unity:

$$z^n = w < \theta$$
$$z = w^{\frac{1}{n}} < \frac{\theta + 2k\pi}{n}$$

When using this, convert the complex number to polar form and place the values into the formula.

7.6 Exponential Form:

$$\begin{aligned} z^n &= e^{jn\theta} & z^{-n} &= e^{-jn\theta} \\ &= \cos(n\theta) + \sin(n\theta)j & &= \cos(n\theta) - \sin(n\theta)j \end{aligned}$$

$$\begin{aligned} z^n + z^{-n} &= 2\cos(n\theta) \\ z^n - z^{-n} &= 2\sin(n\theta)j \end{aligned}$$

$$\cos(n\theta) + \sin(n\theta)j = [\cos(\theta) + \sin(\theta)j]^n$$

For example to prove one of the trig identities:

$$\begin{aligned} [2\cos(\theta)]^2 &= \left(z + \frac{1}{z}\right)^2 \\ 4\cos^2(\theta) &= z^2 + 2 + \frac{1}{z^2} \\ &= 2\cos(2\theta) + 2 \\ \cos^2(\theta) &= \frac{1}{2}[\cos(2\theta) + 1] \end{aligned}$$

7.7 Loci & Regions:

This is just the space used and graphing of the complex numbers. Such as, $(x-a)^2 + (y-b)^2 + (z-c)^2 = r^2$ to graph a circle.